

A Catch-Rematched Throat-Loaded Transit Ansatz for Heavy-Duty Numerical Relativity Testing

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Abstract

This paper presents a stand-alone construction and reduced validation of a throat-loaded, catch-rematched transit ansatz formed from two starting geometries: a smooth Morris–Thorne traversable wormhole throat and a compact ADM warp shell with a Van den Broeck-style capacity factor. The originating question is whether “a compact warp-drive spacetime, especially one with a small external cross-section and large internal volume, [can] be consistently embedded in or propagated through a pre-existing traversable wormhole geometry without creating horizons, singularities, or uncontrollable stress-energy divergences.” The initial combined metric separated spatial capacity, lapse/time-rate support, and transport shift. Successive reduced evaluations then moved capacity and lapse into the wormhole throat complex, gated the transport shift by the same throat support, and finally separated the passenger worldline from the moving field label. The current ansatz uses a passenger/coupling center $X(t)$, caught from an externally superluminal transit speed V to a subluminal exit speed before shift release, and rematches the shift amplitude to the caught passenger velocity $U(t) = \dot{X}(t)$. The resulting radial-axis ADM metric is

$$ds^2 = -T^2 dt^2 + A^2 (dl + \beta^l dt)^2 + A^2 r^2(l) d\Omega^2,$$

with

$$A = \exp(qW \ln C_0), \quad T = \exp(qW \ln(\lambda C_0)), \quad \beta^l = -U(t)E(X)W(l)S(l - X(t)).$$

Reduced validation of the simple catch-rematched baseline found clean passenger worldlines and clean dense g_{tt} maps across the useful core range $V = 1.5, 2.0, 2.5$ and the stretch range $V = 3, 5$. The same baseline shows an off-center edge-shoulder in a deliberately extreme $V = 10$ stress map, which is recorded as a future high-velocity network problem rather than a reason to complicate the baseline. A possible future shift-edge reinforcement, $\beta^l \propto W^{p_\beta}$ with $p_\beta > 1$, is therefore discussed only as a high- V option. The present conclusion is deliberately bounded: the catch-rematched throat-loaded geometry has passed the reduced checks needed to justify real heavy-duty testing, while the decisive open problems remain constraint-quality initial data, dynamical stability, semiclassical/quantum-inequality exposure, and global chronology control.

1 The construction question

The project began with a compositional question rather than a claim of physical realizability. A traversable wormhole already asks spacetime to carry a throat supported by exotic stress-energy. A compact warp-drive construction asks spacetime to carry a moving region whose exterior footprint can be much smaller than its interior spatial capacity. The proposed synthesis asks whether these

burdens can be made to cooperate instead of merely colliding: can the compact warp-side component be embedded in, or propagated through, a pre-existing traversable wormhole geometry while avoiding horizons, singularities, and uncontrolled stress-energy divergences? That question was first written explicitly in the repository note *Compact Warp Shells and Traversable Wormhole Throats* [1].

The starting pair was chosen for structural reasons. The wormhole side was a Morris–Thorne-style throat in proper radial coordinate, because that form keeps the minimum areal radius, flare-out, and redshift freedoms visible [2]. The warp side was expressed in the ADM language emphasized by Bobrick and Martire, because ADM variables separate lapse, shift, and spatial geometry into handles with clear physical roles [3]. The compact-exterior/large-interior motivation came from Van den Broeck’s modification of the Alcubierre construction [4], while the background warning remained the original Alcubierre fact: superluminal warp geometry is an exotic spacetime construction rather than ordinary local motion through flat space [5]. The Krasnikov-tube literature then supplied a second architectural lesson. Everett and Roman stress that a passenger at the center of an Alcubierre-type bubble cannot directly control the outer bubble wall, making infrastructure-like support natural when control of the dangerous region matters [6].

This paper does not attempt to declare the geometry physically buildable. It instead freezes a current reduced ansatz and records why it is mature enough for heavy-duty numerical-relativity testing. The logic is affirmative but bounded. The reduced model has stopped failing cheaply; it has not yet satisfied the Einstein constraints as evolved initial data, nor has it answered the quantum-field-theoretic objections that dominate macroscopic wormhole and warp-drive proposals [7, 8].

2 Base geometry: a Morris–Thorne throat

The wormhole background is written in proper radial coordinate l as

$$ds_{\text{WH}}^2 = -e^{2\Phi(l)} dt^2 + dl^2 + r^2(l) d\Omega^2, \quad d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (1)$$

The throat lies at $l = 0$ with

$$r(0) = r_0 > 0, \quad r'(0) = 0, \quad r''(0) > 0, \quad (2)$$

and the redshift function $\Phi(l)$ is smooth and finite. On a constant- t slice the background spatial metric is

$$q_{ij} dx^i dx^j = dl^2 + r^2(l) d\Omega^2, \quad (3)$$

with scalar curvature

$$^{(3)}R[q] = -\frac{4r''}{r} + \frac{2(1 - (r')^2)}{r^2}. \quad (4)$$

The throat therefore enters every later construction as a real geometric object, not as a coordinate metaphor. It has a local minimum in areal radius and a flare-out contribution that interacts with any capacity gradients placed near it.

The reduced evaluations used the simple smooth choice

$$r(l) = \sqrt{r_0^2 + l^2}, \quad \Phi(l) = 0, \quad (5)$$

which preserves the essential proper-radial throat structure while making the first scans reproducible. This reduction is intentionally conservative in one sense and incomplete in another. It exposes how the support, shift, and release profiles interact near a throat; it does not yet optimize the full wormhole redshift or shape-function freedom.

3 Base geometry: ADM warp shell and spatial capacity

The warp-side construction begins with the ADM metric

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt) (dx^j + \beta^j dt), \quad (6)$$

where α is the lapse, β^i is the shift, and γ_{ij} is the spatial metric. The first combined ansatz placed a compact shell profile $S(\rho)$ around a center $L(t)$ and used

$$\alpha = e^{\Phi(l)} T(\rho), \quad \gamma_{ij} = A^2(\rho) q_{ij}, \quad \beta^i = -\dot{L}(t) S(\rho) n^i. \quad (7)$$

The Van den Broeck-style capacity factor was represented by a positive exponential interpolation,

$$A(ho) = \exp(S(\rho) \ln C_0), \quad C_0 > 1, \quad (8)$$

with analogous positive interpolation for T . In the radial-axis reduction, this produced

$$ds^2 = -e^{2\Phi(l)} T^2 dt^2 + A^2 \left(dl - \dot{L}(t) S dt \right)^2 + A^2 r^2(l) d\Omega^2. \quad (9)$$

The determinant is fixed by ADM form,

$$\det g_{\mu\nu} = -\alpha^2 \det \gamma_{ij} = -e^{2\Phi} T^2 A^6 r^4 \sin^2 \theta, \quad (10)$$

so positive finite A, T, r preserve nondegenerate Lorentzian signature away from ordinary polar-coordinate degeneracy.

The first ansatz exposed a problem of responsibility. A moving compact shell can carry capacity, lapse, and shift, but if the high-burden parts move with the passenger bubble, the bubble is being asked to carry the hard geometry itself. The later throat-loaded branch moved the high-burden support into the wormhole complex.

4 Throat-loaded support and the gated-shift result

The throat-loaded refinement places capacity and lapse under a throat support profile $W(l)$ and a release profile $q(L)$,

$$A(l, L) = \exp(q(L) W(l) \ln C_0), \quad T(l, L) = \exp(q(L) W(l) \ln(\lambda C_0)). \quad (11)$$

Here $W(l)$ is a smooth throat support function and q controls the release of throat capacity/lapse as the passenger approaches the exit layer. The transport/coupling profile remains localized near the passenger center. The decisive comparison was between an independent passenger shift,

$$\beta^l = -V E(L) S_{\text{pass}}(l - L), \quad (12)$$

and a throat-gated shift,

$$\beta^l = -V E(L) W(l) S_{\text{pass}}(l - L). \quad (13)$$

The repository report *Throat-Loaded Transit Architecture: Gated-Shift Reduced Evaluation* records the resulting architectural selection: capacity, lapse, and shift should behave as one managed throat system [9]. In the 336-configuration reduced pass through $V = 2.0$, the independent passenger-shift comparison entered lapse-shift balance crossings in many cases, while the throat-gated family

produced no traversal or exit configurations with positive g_{tt} points and kept all 336 traversal-and-exit cases clean in the sampled ladder.

This is exactly what the radial metric predicts. The lapse-shift monitor is

$$g_{tt} = -T^2 + A^2 V^2 E^2 W^2 S^2. \quad (14)$$

An independent shift can extend transport support outside the region still carrying throat lapse and capacity. A throat-gated shift suppresses transport where the throat support W is weak. The first robust design rule is therefore geometric rather than cosmetic:

$$\text{supp}(\beta^l) \subseteq \text{supp}(A, T). \quad (15)$$

The release ordering sharpened the same idea. Let E control transport support and q control capacity/lapse support. A shift-first choreography uses

$$L_\beta < L_q, \quad (16)$$

so that $E \rightarrow 0$ before $q \rightarrow 0$. In words, transport fades before the throat relaxes. Profile robustness evaluations confirmed that this ordering is not a decorative preference. Mixed synchronized release can fail when the effective E profile remains high while q has already dropped. The failure is then the expected inequality violation,

$$A V E W S > T. \quad (17)$$

The result turned an informal control intuition into a design constraint: support must outlive transport.

5 The superluminal worldline problem

The throat-gated metric fixes the local ADM causal monitor, but superluminal passenger transport requires a second question. The earlier reduced family used $L(t) = Vt$ as the moving center. If this curve is interpreted as the passenger trajectory, then the norm of the center curve is approximately

$$\frac{ds^2}{dt^2} = -T^2 + A^2 (V - V E W S)^2. \quad (18)$$

Inside full support, $E W S \approx 1$ and the shift cancels the externally superluminal coordinate motion. During release, $E W S \rightarrow 0$ and the same curve approaches

$$\frac{ds^2}{dt^2} \approx -1 + V^2. \quad (19)$$

For $V > 1$ this is spacelike. This was the first genuinely physical early-break result of the source/worldline anatomy pass. The field profile can be regular while the passenger interpretation is incomplete.

The remedy is the one the architecture already suggested: a catch/rematching phase. The moving field label and the passenger/coupling worldline should be separated. The passenger center is denoted $X(t)$ and has velocity

$$U(t) = \dot{X}(t). \quad (20)$$

It is caught smoothly from a transit speed V to a subluminal exit speed v_{exit} before shift support fades. A representative catch law is

$$U(t) = v_{\text{exit}} + (V - v_{\text{exit}})C(t), \quad (21)$$

where C falls from 1 to 0 inside the throat-supported region. The ordering becomes

$$L_{\text{catch}} < L_\beta < L_q. \quad (22)$$

The physical reading is direct: catch the passenger first, fade shift second, relax throat third.

6 The catch-rematched ansatz

A catch applied only to the passenger is not enough at high V if the shift field remains tied to the original cruise speed. The shift must be caught with the passenger. The present baseline ansatz is therefore

$$\boxed{\beta^l(l, t) = -U(t)E(X(t))W(l)S(l - X(t))}. \quad (23)$$

The full reduced radial metric is

$$\boxed{ds^2 = -T^2 dt^2 + A^2 \left(dl + \beta^l dt \right)^2 + A^2 r^2(l) d\Omega^2}, \quad (24)$$

with A and T given by Eq. (11). The passenger/coupling profile is centered on $X(t)$, and the shift amplitude follows $U(t)$ rather than the original V once catch begins.

This choice resolves the earlier worldline failure at the level of the reduced metric. Along the passenger center $l = X(t)$, the timelike condition is

$$d\tau^2 = \left[T^2 - A^2 \left(U + \beta^l \right)^2 \right] dt^2 > 0. \quad (25)$$

With Eq. (23), the dangerous relative velocity is

$$v_{\text{rel}} = \frac{A}{T} (U - U E W S), \quad (26)$$

so the fully supported region cancels the motion and the exit region is already subluminal because $U \rightarrow v_{\text{exit}} < 1$ before $E \rightarrow 0$.

The reduced source is still read off from Einstein's equation,

$$T_{\mu\nu} = \frac{1}{8\pi} G_{\mu\nu}, \quad (27)$$

rather than supplied by a matter model. That keeps the present paper at the prescribed-geometry stage. It is exactly why the next heavy-duty step is constraint-quality initial data and dynamical evolution.

7 Reduced diagnostic suite

The reduced evaluations used invariant or frame-defined diagnostics tied to the ADM decomposition. The local causal monitor was g_{tt} ,

$$g_{tt} = -T^2 + A^2(\beta^l)^2. \quad (28)$$

The direct passenger condition was the norm of the passenger center curve,

$$\mathcal{N}_X(t) = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu, \quad x^\mu(t) = (t, X(t), \pi/2, 0). \quad (29)$$

The reduced curvature diagnostics included R , $R_{\mu\nu}R^{\mu\nu}$, the Kretschmann scalar, reduced null Einstein projections $G_{\mu\nu}k_{\pm}^{\mu}k_{\pm}^{\nu}$, reduced null expansions $\theta_{\pm}$, and ADM-normal orthonormal tidal proxies. The burden proxy used in the earlier throat-gated evaluation integrated a positive combination of curvature magnitude, square-root Kretschmann magnitude, negative null projection parts, and tidal magnitudes over throat and passenger regions. Later source anatomy decomposed the Einstein tensor in an ADM-normal orthonormal frame into energy-density, pressure/tension, and radial flux proxies.

The final proper reduced validation froze the simple catch-rematched baseline and separated the target velocities into three classes. The core range was $V = 1.5, 2.0, 2.5$. The stretch range was $V = 3, 5$. The deliberately extreme stress point was $V = 10$. This hierarchy matters. The useful engineering question is first whether the current system is coherent in the already valuable $V \sim 1.5$ – 2.5 range and then whether it continues into a strong stretch range. The $V = 10$ point is a future high-velocity network stress case, not the baseline design center.

8 Final reduced validation results

The final reduced validation package [10] contains dense worldline sampling, convergence-selected worldline checks, dense metric/source maps near catch/release, focused timing-basin perturbations, boundary failure cases, and an initial-data-readiness proxy. The central result is summarized in Table 1. The simple catch-rematched baseline passes the core and stretch reduced checks. It develops an off-center $g_{tt} > 0$ shoulder only in the extreme $V = 10$ stress map.

Table 1: Final reduced validation hierarchy for the simple catch-rematched baseline. “Clean” means no passenger worldline timelike failures and no positive- g_{tt} points in the corresponding dense reduced metric maps.

Velocity range	Speeds	Reduced result
Core	$V = 1.5, 2.0, 2.5$	Clean passenger worldlines and clean dense g_{tt} maps
Stretch	$V = 3, 5$	Clean passenger worldlines and clean dense g_{tt} maps
Extreme stress	$V = 10$	Passenger worldline clean; off-center dense-map $g_{tt} > 0$ edge shoulder

The result is stronger than the original $V \leq 2$ scan in the right way. The original scan demonstrated that throat gating and shift-first release stabilize the ADM causal monitor through $V = 2$ for the throat-loaded family. The catch-rematched scan demonstrates that superluminal passenger interpretation no longer fails cheaply when a physical catch phase is included. The present baseline is coherent through $V = 5$ in reduced tests.

The final timing-basin scans also clarify how the ansatz fails. Boundary cases appear when catch occurs too late, too close to a weakening throat-support edge, or with too little separation between catch and shift release. The failures are therefore not patternless. They follow the hierarchy already built into the construction:

$$\text{catch} \longrightarrow \text{fade shift} \longrightarrow \text{relax throat}. \quad (30)$$

A design that violates this ordering eventually allows relative motion or shift support to outlive the geometry that keeps it causal.

9 Why the geometry is ready for heavy-duty testing

The case for heavy-duty testing rests on the nature of the failures found so far. The early moving-shell ansatz did not merely accumulate arbitrary numerical artifacts; it revealed that capacity, lapse, and shift have to be organized by the throat. The independent-shift branch then failed in the expected way, with transport extending beyond support. Profile robustness did not find an unexplained dependency on one special function; it found that effective release ordering must be preserved. Source/worldline anatomy then found the superluminal center-curve problem, and catch/rematching fixed that problem without breaking the reduced metric in the core or stretch range.

This sequence matters because it turns reduced failures into design principles. The active throat complex carries the high-burden geometric functions. Transport remains throat-gated. The passenger is caught before shift release. The shift amplitude is rematched to the caught passenger velocity. The throat relaxes last. These principles are specific enough to define an initial-data target.

The current ansatz is therefore ready for heavy-duty testing in a precise sense. It is ready to be tested as a candidate geometry for constraint-quality initial data and numerical evolution. It is not ready to be treated as a physically realized transit system. The remaining physics is exactly the difficult physics that requires stronger tools: the Hamiltonian and momentum constraints, dynamical stability, quantum inequalities, source modeling, and global chronology.

10 Initial-data target

On a constant- t slice the current ansatz gives

$$\gamma_{ij} = A^2 q_{ij}, \quad \alpha = T, \quad \beta^i = (\beta^l, 0, 0). \quad (31)$$

The extrinsic curvature target is

$$K_{ij} = \frac{1}{2\alpha} (D_i \beta_j + D_j \beta_i - \partial_t \gamma_{ij}), \quad (32)$$

where D_i is compatible with γ_{ij} . The constraint diagnostics are then

$$\mathcal{H} = {}^{(3)}R + K^2 - K_{ij} K^{ij} - 16\pi\rho, \quad (33)$$

$$\mathcal{M}^i = D_j (K^{ij} - \gamma^{ij} K) - 8\pi j^i. \quad (34)$$

At the prescribed-metric stage, ρ and j^i can be read from the Einstein tensor. A real initial-data construction must decide whether to accept an effective source, solve for compatible matter content, or reformulate part of the geometry as freely specifiable data in an elliptic constraint solve.

The recommended heavy-duty path starts with axisymmetric or radial-plus-angular initial-data diagnostics at $V = 2$ or $V = 2.5$, then repeats for $V = 5$ after the lower-speed case behaves. Full $3 + 1$ evolution belongs after this constraint-quality pass, not before it.

11 Engineering interpretation

The architecture implied by the reduced ansatz is infrastructure-first. The passenger object does not carry the entire exotic geometry as a self-contained warp vehicle. The wormhole throat complex is the active machine. It manages capacity, lapse, shift, release timing, and catch. The passenger/coupling region is admitted, carried while supported, caught to a timelike exterior state, and released after transport has faded.

This interpretation naturally favors gates, nodes, and segmented networks rather than a single monolithic corridor through ordinary exterior space. For a multi-light-year trip, the exotic support need not mean a literal physical pipe of exotic material laid through the external interstellar distance. In a wormhole-like mouth/throat architecture, the support is tied to the geometry that identifies the mouths and throat, with transition and catch layers localized to the managed complexes. If the architecture is extended into a network, each link or node has to obey its own catch–shift–relax ordering, and the network as a whole must obey chronology constraints. Morris, Thorne, and Yurtsever’s time-machine warning is therefore not peripheral; it is an engineering constraint for any separated-mouth system [11].

Throughput is also an infrastructure question. The current reduced ansatz describes one active passenger/coupling packet per managed transit/catch channel. Sequential trips through a single local throat gate are the safe baseline. Multiple vessels require multiple separated packets, parallel channels, or a block-signaled network in which no overlapping shift/catch profiles violate the shared lapse-shift margin.

12 Future high-velocity option: reinforced throat-edge shift gating

The final validation deliberately did not alter the simple baseline in response to the $V = 10$ stress shoulder. The baseline passes through $V = 5$, and $V = 10$ belongs to a future high-velocity network regime. Still, the $V = 10$ map identified a useful optional path if later work targets much higher V .

The edge-shoulder occurs where W is partial, S remains appreciable, and the lapse ratio

$$\frac{T}{A} = \lambda^{qW} \quad (35)$$

has weakened enough that the shift contribution can locally dominate. A natural future modification is to gate the shift more strongly by the throat support,

$$\beta^l = -U(t)E(X)W^{p_\beta}(l)S(l - X(t)), \quad p_\beta > 1. \quad (36)$$

This is recorded as a future high- V option because it directly suppresses the shift shoulder in partial-support regions while leaving the fully supported center almost unchanged. It is not part of the present baseline. It should only be promoted if the simple ansatz fails in a velocity range that the engineering target actually requires.

13 Physical limitations

All reduced successes in this paper remain inside classical prescribed geometry. The source is not a known material; it is the Einstein tensor divided by 8π . Negative energy density and null-energy-condition violation are expected because both traversable wormholes and superluminal warp constructions live in that family. Quantum inequalities and semiclassical backreaction remain the strongest theoretical threats. Ford and Roman’s wormhole constraints and Pfenning and Ford’s warp-drive analysis are direct warnings that localized exotic support can become physically unattainable at macroscopic scale [7, 8].

The reduced tidal and source diagnostics should also be read in scale-aware terms. Dimensionless tidal peaks do not automatically specify a physical size. Once a length scale L_0 is assigned, curvature and tidal accelerations scale like L_0^{-2} . A compact laboratory-scale implementation would therefore face vastly larger physical tides than an astrophysically scaled complex. The current paper does not solve that scale problem. It identifies a geometry coherent enough that the scale problem is worth asking with stronger methods.

14 Conclusion

The catch-rematched throat-loaded transit ansatz is the current coherent endpoint of the reduced construction. It began with a compact warp shell embedded in a traversable wormhole background, moved capacity and lapse into the throat, gated shift by throat support, enforced shift-first release, separated the passenger worldline from the old moving field label, and rematched the shift amplitude to the caught passenger velocity. The final baseline is

$$\begin{aligned} A(l, t) &= \exp(q(t)W(l) \ln C_0), \\ T(l, t) &= \exp(q(t)W(l) \ln(\lambda C_0)), \\ \beta^l(l, t) &= -U(t)E(X(t))W(l)S(l - X(t)), \end{aligned}$$

with the operational ordering

$$\text{catch } U(t) \rightarrow \text{fade } E \rightarrow \text{relax } q. \quad (37)$$

The simple baseline passes reduced validation through the core $V = 1.5$ – 2.5 range and the stretch $V = 3$ – 5 range. It is therefore ready for constraint-quality initial-data construction and heavy-duty numerical relativity testing. The next work should build and solve the corresponding 3+1 initial-data problem, starting at $V = 2$ or $V = 2.5$, before attempting full dynamical evolution or high-velocity network variants.

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